The documentation and process conversion measures necessary to comply with this revision shall be completed by 26 November 2014.

INCH-POUND

MIL-PRF-19500/544H
SUPERSEDING
26 August 2014
SUPERSEDING
MIL-PRF-19500/544G
12 April 2008

PERFORMANCE SPECIFICATION SHEET

* SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER,
TYPES 2N5152, 2N5154, 2N5152L, 2N5154L, 2N5152U3, 2N5154U3, JAN, JANTX, JANTXV, JANTXVF, JANTXVR, JANS, JANSM, JANSD, JANSP, JANSL, JANSR, JANSF, JANSG, JANSH, JANHCB, JANHCD, JANHCE, JANKCB, JANKCBF, JANKCD, JANKCE, JANKCDM, JANKCDD, JANKCDP, JANKCDL, JANKCDR, JANKCDF, JANKCDG, AND JANKCDH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, power transistors for use in high-speed power-switching applications. Four levels of product assurance are provided for each encapsulated device type and two levels of product assurance are provided for each unencapsulated device type as specified in MIL-PRF-19500. RHA level designators “M”, “D”, “P“, “L,” “R”, “F”, “G”, and “H” are appended to the device prefix to identify devices, which have passed RHA requirements.

1.2 Physical dimensions. See figure 1 (similar to TO-5 and TO-39), figures 2, 3, and 4 (die dimensions), and figure 5 (U3).

* 1.3 Maximum ratings. Unless otherwise specified T_A = +25°C.

Type	I_C	V_{CBO}	V_{CEO}	V_{EBO}	T_{stg} and T_J
	A dc	V dc	V dc	V dc	°C
2N5152	2	100	80	5.5	
2N5152L	2	100	80	5.5	
2N5154	2	100	80	5.5	-65 to
2N5154L	2	100	80	5.5	
2N5152U3	2	100	80	5.5	+200
2N5154U3	2	100	80	5.5	

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 5961

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* 1.3 Maximum ratings. Unless otherwise specified T_A = +25°C. - Continued.

Type	P_T T_A = +25°C	P_T T_C = +25°C	P_T T_{SP} = +25°C	I_C (1)	$R_{\theta JC}$	$R_{\theta JA}$	Reverse pulse energy (2)
	W (3)	W (3)	W	A dc	°C/W (4)	°C/W (4)	mJ
2N5152	1	10	N/A	10	10	175	15
2N5152L	1	10	N/A	10	10	175	15
2N5154	1	10	N/A	10	10	175	15
2N5154L	1	10	N/A	10	10	175	15
2N5152U3	N/A	100	1 (3)	10	1.7	170 (5)	15
2N5154U3	N/A	100	1 (3)	10	1.7	170 (5)	15

* (1) This collector current value applies for Pw ≤ 8.3 ms, duty cycle ≤ 1 percent (page 1).
* (2) This rating is based on the capability of the transistors to operate safely in the unclamped inductive load energy test circuit, see subgroup 5 of the group A inspection table.
(3) For derating, see figures 6, 7, 8, and 9.
(4) For thermal impedance curves, see figures 10, 11, and 12.
(5) Mounted on an FR4 printed circuit board.

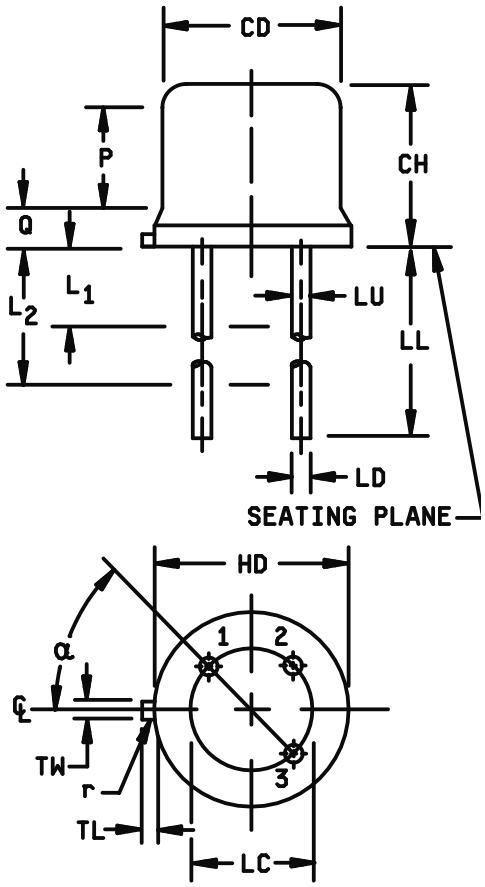
* 1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25°C.

Limits	h_{FE2} (1) V_{CE} = 5 V I_C = 2.5 A		$\lvert h_{fe} \rvert$ V_{CE} = 5 V I_C = 500 mA dc f = 10 MHz		$V_{BE(sat)2}$ (1) I_C = 5 A dc I_B = 500 mA dc	$V_{CE(sat)2}$ (1) I_C = 5 A dc I_B = 500 mA dc	C_{obo} V_{CB} = 10 V dc I_E = 0 f = 1 Mhz
	2N5152 (2)	2N5154 (2)	2N5152 (2)	2N5154 (2)			
					Vdc	Vdc	pF
Min	30	70	6	7			
Max (TO-5, TO-39)	90	200			2.2	1.5	250
Max (U3)	90	200			2.2	1.5	250

(1) Pulsed see 4.5.1.
(2) The limits specified apply to all package outlines unless otherwise stated.

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	6
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		7
LD	.016	.019	0.41	0.48	8,9
LL	See note 14				
LU	.016	.019	0.41	0.48	8,9
L_1		.050		1.27	8,9
L_2	.250		6.35		8,9
P	.100		2.54		7
Q		.030		0.76	5
TL	.029	.045	0.74	1.14	3,4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45° TP		45° TP		7
1, 2, 10, 12, 13, 14					



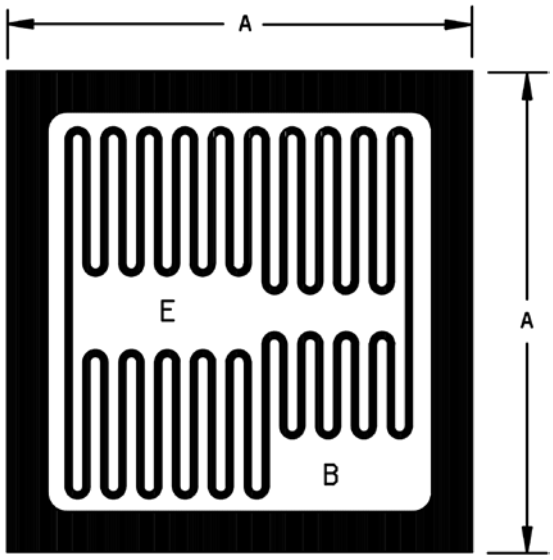
NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by gauging procedure.
8. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in and beyond LL minimum.
9. All three leads.
10. The collector shall be internally connected to the case.
11. Dimension r (radius) applies to both inside corners of tab.
12. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.
13. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

* 14. For L-suffix devices (TO-5), dimension LL = 1.5 inches (38.10 mm) min. and 1.75 inches (44.45 mm) max. For no suffix types (TO-39), dimension LL = .5 inch (12.70 mm) min. and .750 inch (19.05 mm) max.

* FIGURE 1. Physical dimensions (similar to TO-5 and TO-39).

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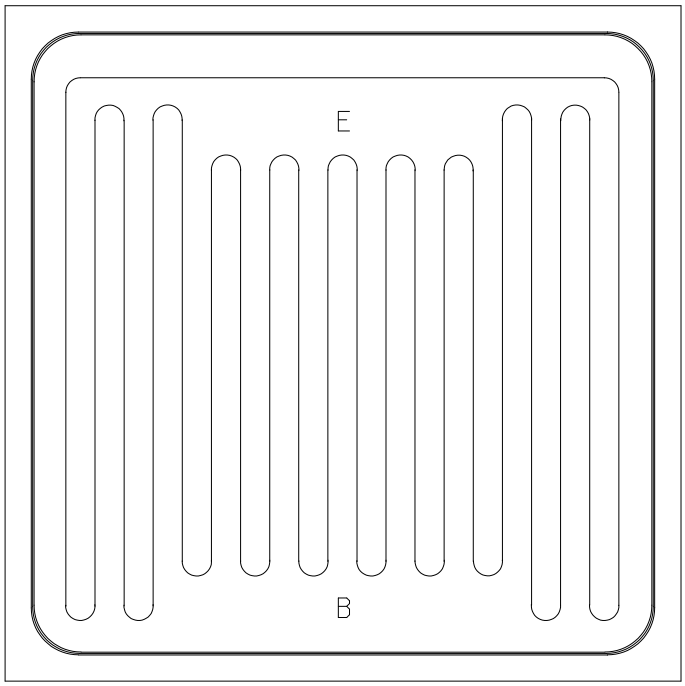


Dimensions				
LTR	Inches		Millimeters	
	Min	Max	Min	Max
A	.095	.105	2.41	2.66

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. The physical characteristics of the die are:
   Thickness: .0078 (0.198 mm) nominal, tolerance is ±.005 (0.13 mm).
   Top metal: Aluminum, 25,000 Å minimum, 33,000 Å nominal.
   Back metal: Gold 1,500 Å minimum, 2,500 Å nominal.
   Back side: Collector.
   Bonding pad: .012 (0.305 mm) min. x .030 (0.761 mm) minimum.

FIGURE 2. JANHC and JANKC (B-version) die dimensions.

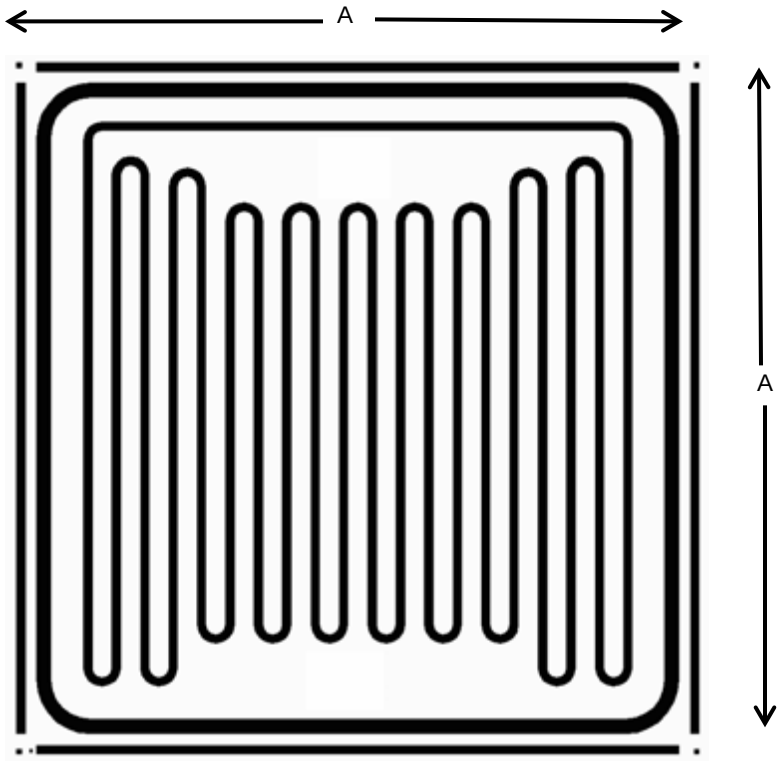
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1.	Die size	.120 inch (3.05 mm) x .120 inch (3.05 mm) ±.002 inch (±0.05 mm).
2.	Die thickness	.014 inch (0.35 mm) ±.0015 inch nominal (±0.04 mm).
3.	Top metal	Aluminum, 54,000Å minimum, 60,000Å nominal.
4.	Back metal	A. Al/Ti/Ni/Ag, 10,000Å minimum, 12,500Å nominal. B. Gold 6,000Å minimum, 8,000Å nominal.
5.	Backside	Collector
6.	Bonding pad	B = .060 x .012 inch (1.5 mm x 0.30 mm). E = .050 x .012 inch (1.27 mm x 0.30 mm).

FIGURE 3. JANHC and JANKC (D-version) die dimensions.

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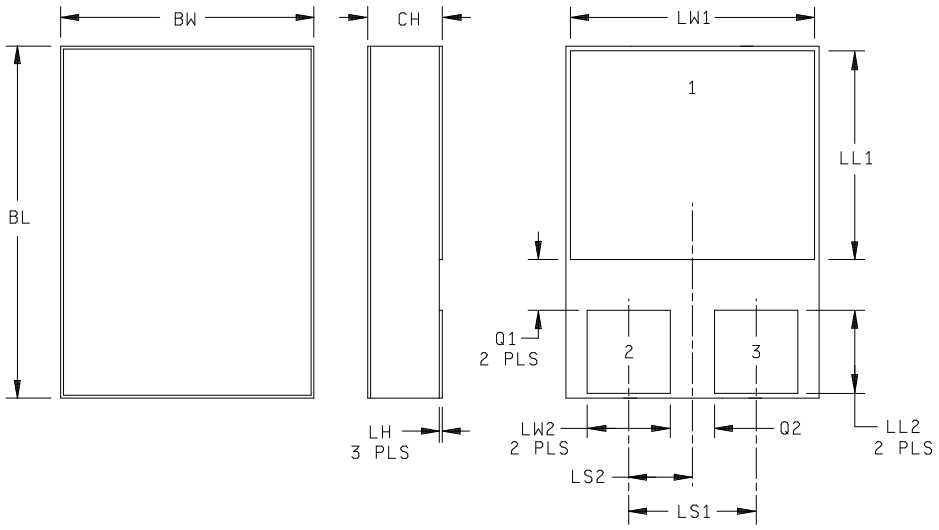


Dimensions				
LTR	Inches		Millimeters	
	Min	Max	Min	Max
A	.118	.122	3.0	3.1

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. The physical characteristics of the die are:
   - Thickness: .014 inch (0.35 mm) nominal, tolerance is ±.0015 (0.04 mm).
   - Top metal: Aluminum, 54,000 Å minimum, 60,000 Å nominal.
   - Back metal: Gold 6,400 Å minimum, 8,000 Å nominal.
   - Back side: Collector.
   - Bonding pad: B = .060 x .012 inch (1.5 mm x 0.30 mm)
     E = .050 x 0.12 inch (1.27 mm x 0.30 mm)

* FIGURE 4. JANHC and JANKC (E-version) die dimensions.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.395	.405	10.04	10.28
BW	.291	.301	7.40	7.64
CH	.1085	.1205	2.76	3.06
LH	.010	.020	0.25	0.51
LW1	.281	.291	7.14	7.41
LW2	.090	.100	2.29	2.54
LL1	.220	.230	5.59	5.84
LL2	.115	.125	2.93	3.17
LS1	.150 BSC		3.81 BSC	
LS2	.075 BSC		1.91 BSC	
Q1	.030		0.762	
Q2	.030		0.762	
TERM 1	Collector			
TERM 2	Base			
TERM 3	Emitter			

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 5. Physical dimensions and configuration for surface mount (U3).

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (similar to TO-5 and TO-39), and figures 2, 3, and 4 (die dimensions) for JANHC and JANKC, and figure 5 (U3).

3.4.1 Current density. Current density of internal conductors shall be as specified in MIL-PRF-19500.

3.4.2 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

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3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements, PIN designators, and test levels shall be as defined in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500. The radiation hardened designator M, D, P, L, R, F, G, or H shall immediately precede (or replace) the device “2N” identifier (depending upon degree of abbreviation required).

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, III, and IV).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement	
	JANS levels	JANTX and JANTXV levels
(1) 3c	Thermal impedance, method 3131 of MIL-STD-750. See 4.3.3.	Thermal Impedance, method 3131 of MIL-STD-750. See 4.3.3.
9	I_{CES1} and h_{FE2}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or100 nA dc, whichever is greater. Δh_{FE2} = ±20 percent.	I_{CES1} and h_{FE2}
12	See 4.3.2	See 4.3.2
13	Subgroup 2 and 3 of table I herein; ΔI_{CES1} = 100 percent of initial value or 100 nA dc, whichever is greater. Δh_{FE2} = ±20 percent.	Subgroup 2 of table I herein; ΔI_{CES1} = 100 percent of initial value or 100 nA dc, whichever is greater. Δh_{FE2} = ±20 percent.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, " Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to the device to achieve T_J = +175°C minimum using a minimum P_D = 75 percent of P_T maximum, T_A ambient rated as defined in 1.3 herein.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, $t_{MD,}$ (and V_C where appropriate). The $Z_{\theta JX}$ limit used in screen 3c of 4.3 herein shall comply with the thermal impedance graph in figure 9, 10, and 11 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table IV herein.

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4.4.2.1 Group B inspection (JANS) table E-VIA of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	V_{CB} = 10 V dc.
B5	1027	V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table E-VIA, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve a T_J = +225°C minimum.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new assembly lot option is exercised, the failed assembly lot shall be scrapped.

Step	Method	Conditions
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and 4.4.3.1 (JANS), and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and table IV herein.

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4.4.3.1 Group C inspection, table E-VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E (not applicable to U3 devices).
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and applied thermal impedance curves.
C6	1026	1,000 hours, V_{CB} = 10 V dc, power and ambient temperature shall be applied to the device to achieve T_J = +150°C minimum, and minimum power dissipation of 75 percent of max rated P_T (see 1.3 herein); n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Group C inspection, table E-VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E (not applicable to U3 devices).
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3).
C6	1037	Not applicable.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2 herein), except group D, subgroup 2 may be performed separate from other subgroups. Group D inspection may also be performed ahead of the screening lot using die selected in accordance with MIL-PRF-19500 and related documents. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
*	Subgroup 1 2/						
	Visual and mechanical examination 3/	2071					
	Solderability 3/ 4/	2026					
	Resistance to solvents 3/ 4/ 5/	1022					
	Temp cycling 3/ 4/	1051	Test condition C, 25 cycles				
	Hermetic seal 4/ 6/	1071					
	Fine leak Gross leak						
	Electrical measurements 4/		Group A, subgroup 2				
	Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hours or T_A = +300°C at t = 2 hours				
	Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
	Subgroup 2						
	Thermal impedance 7/	3131	See 4.3.3	$Z_{\theta JX}$			°C/W
	Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 100 mA dc; I_B = 0, pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
	Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 60 V dc; V_{BE} = 0	I_{CES1}		1.0	µA dc
	Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 100 V dc; V_{BE} = 0	I_{CES2}		1.0	mA dc
	Collector to emitter cutoff current	3041	Bias condition D, V_{CE} = 40 V dc, I_B = 0	I_{CEO}		50	µA dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4 V dc, I_C = 0	I_{EBO1}		1.0	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 5.5 V dc, I_C = 0	I_{EBO2}		1.0	mA dc
Forward current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 50 mA dc	h_{FE1}			
2N5152, L, and U3 2/ 2N5154, L, and U3				20 50		
Forward current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 2.5 A dc, pulsed (see 4.5.1)	h_{FE2}			
2N5152, L, and U3 2/ 2N5154, L, and U3				30 70	90 200	
Forward current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 5 A dc, pulsed (see 4.5.1)	h_{FE3}			
2N5152, L, and U3 2/ 2N5154, L, and U3				20 40		
Base-emitter voltage (nonsaturated)	3066	Test condition B, V_{CE} = 5 V dc, I_C = 2.5 A dc, pulsed (see 4.5.1)	V_{BE}		1.45	V dc
Base-emitter saturation voltage	3066	Test condition A, I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.45	V dc
Base-emitter saturation voltage	3066	Test condition A, I_C = 5 A dc, I_B = 500 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.2	V dc
Collector-emitter saturation voltage	3071	I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.75	V dc
Collector-emitter saturation voltage	3071	I_C = 5 A dc, I_B = 500 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.5	V dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_C = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{CE} = 60 V dc, V_{BE} = -2 V dc	I_{CEX}		25	µA dc
Low temperature operation		T_C = -55°C				
Forward - current transfer ratio 2N5152, L, and U3 2/ 2N5154, L, and U3	3076	V_{CE} = 5 V dc, I_C = 2.5 A dc, pulsed (see 4.5.1)	h_{FE4}	 15 25		
Subgroup 4						
Common-emitter, small-signal, short-circuit, forward-current transfer ratio	3206	V_{CE} = 5 V dc I_C = 100 mA dc, f = 1 kHz	h_{fe}			
2N5152, L, and U3 2/ 2N5154, L, and U3				20 50		
Magnitude of common-emitter, small-signal short-circuit, forward-current, transfer ratio	3306	V_{CE} = 5 V dc, I_C = 500 mA dc, f = 10 MHz	$\lvert h_{fe} \rvert$			
2N5152, L, and U3 2/ 2N5154, L, and U3				6 7		
Open-circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, f = 1 MHz	C_{obo}		250	pF
Switching time		I_C = 5 A dc, I_{B1} = 500 mA dc	t_{on}		0.5	µs
		I_{B2} = -500 mA dc	t_s		1.4	µs
		$V_{BE(off)}$ = 3.7 V dc	t_f		0.5	µs
		R_L = 6Ω; (see figure 13)	t_{off}		1.5	µs

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (dc)	3051	Pre-pulse condition for each test: T_C = +25°C Pulse condition for each test t_p = 1 sec. 1 cycle, T_C = +25°C, (see figure 14)				
Test # 1		V_{CE} = 5.0 V dc, I_C = 2 A dc for TO-39, TO-5, and U3				
Test # 2		V_{CE} = 32 V dc, I_C = 310 mA dc				
Test # 3		V_{CE} = 80 V dc, I_C = 12.5 mA dc				
Safe operating area (unclamped inductive)		T_C = +25°C, R_{BB1} = 10Ω R_{BB2} = 100Ω, L = 0.3 mH, RL = 0.1Ω, V_{CC} = 10 V dc, V_{BB1} = 10 V dc, V_{BB2} = 4 V dc, I_{CM} = 10 A dc, (see figure 15)				
End-point electrical measurements		See table I, subgroup 2				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in subgroup 1, of table II, shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ Hermetic seal test is an end-point to temperature cycling in addition to electrical measurements.
7/ This test required for the following end-point measurements only:
Group B, step 1 of 4.4.2.2 herein (JAN, JANTX, and JANTXV).
Group B, subgroups 3, 4, and 5 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1 and 2.

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron irradiation	1017	Neutron exposure V_{CES} = 0V				
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 100 mA dc; I_B = 0, pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 60 V dc; V_{BE} = 0	I_{CES1}		2.0	µA dc
Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 100 V dc; V_{BE} = 0	I_{CES2}		2.0	mA dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 40 V dc, I_B = 0	I_{CEO}		100	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4 V dc, I_C = 0	I_{EBO1}		2.0	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 5.5 V dc, I_C = 0	I_{EBO2}		2.0	mA dc
Forward-current transfer ratio 2N5152 2N5154	3076	V_{CE} = 5 V dc; I_C = 50 mA dc	[h_{FE1}] 5/	 [10] [25]		
Forward-current transfer ratio 2N5152 2N5154	3076	V_{CE} = 5 V dc; I_C = 2.5 A dc, Pulsed	[h_{FE2}] 5/	 [15] [35]	 90 200	
Forward-current transfer ratio 2N5152 2N5154	3076	V_{CE} = 5 V dc; I_C = 5 A dc, pulsed	[h_{FE3}] 5/	 [10] [20]		
Base-emitter voltage (nonsaturated)	3066	Test condition B, V_{CE} = 5 V dc, I_C = 2.5 A dc, pulsed (see 4.5.1)	V_{BE}		1.45	V dc
Base-emitter saturation voltage	3066	Test condition A, I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.67	V dc
Base-emitter saturation voltage	3066	Test condition A, I_C = 5 A dc, I_B = 500 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.53	V dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 - Continued.						
Collector-emitter saturation voltage	3071	I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.86	V dc
Collector-emitter saturation voltage	3071	I_C = 5 A dc, I_B = 500 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.73	V dc
Subgroup 2						
Total dose irradiation	1019	Gamma exposure V_{CES} = 64 V				
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 100 mA dc; I_B = 0, pulsed (see 4.5.1)	$V_{(BR)CEO}$	80		V dc
Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 60 V dc; V_{BE} = 0	I_{CES1}		2.0	µA dc
Collector to emitter cutoff current	3041	Bias condition C, V_{CE} = 100 V dc; V_{BE} = 0	I_{CES2}		2.0	mA dc
Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 40 V dc, I_B = 0	I_{CEO}		100	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 4 V dc, I_C = 0	I_{EBO1}		2.0	µA dc
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 5.5 V dc, I_C = 0	I_{EBO2}		2.0	mA dc
Forward-current transfer ratio 2N5152 2N5154	3076	V_{CE} = 5 V dc; I_C = 50 mA dc	[h_{FE1}] 5/	 [10] [25]		
Forward-current transfer ratio 2N5152 2N5154	3076	V_{CE} = 5 V dc; I_C = 2.5 A dc	[h_{FE2}] 5/	 [15] [35]	 90 200	
Forward-current transfer ratio 2N5152 2N5154	3076	V_{CE} = 5 V dc; I_C = 5 A dc. pulsed	[h_{FE3}] 5/	 [10] [20]		

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Base-emitter voltage (nonsaturated)	3066	Test condition B, V_{CE} = 5 V dc, I_C = 2.5 A dc, pulsed (see 4.5.1)	V_{BE}		1.45	V dc
Base-emitter saturation voltage	3066	Test condition A, I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.67	V dc
Base-emitter saturation voltage	3066	Test condition A, I_C = 5 A dc, I_B = 500 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.53	V dc
Collector-emitter saturation voltage	3071	I_C = 2.5 A dc, I_B = 250 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.86	V dc
Collector-emitter saturation voltage	3071	I_C = 5 A dc, I_B = 500 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.73	V dc

1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to all device types unless otherwise noted.
4/ Subgroup 1 is an optional test and must be specified on the contract when required.
5/ See method 1019 of MIL-STD-750 for how to determine [h_{FE}] by first calculating the delta ($1/h_{FE}$) from the pre- and post-radiation h_{FE}. Notice that [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) - for qualification only.

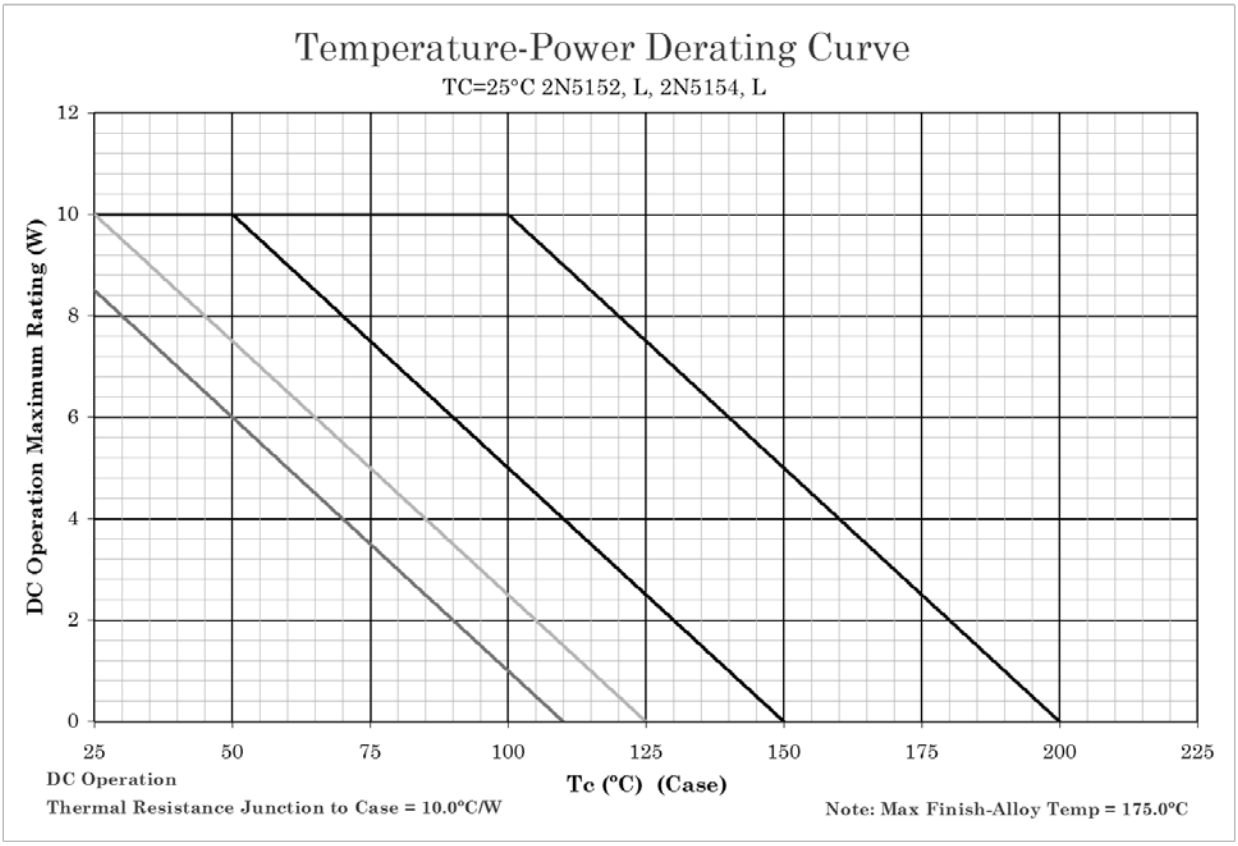
Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal			
Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 and table IV herein	
Subgroup 2			45 devices c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles	
Electrical measurements		See table I, subgroup 2 and table IV herein	
Subgroup 4			
Thermal impedance		See MIL-PRF-19500, table E-IX, group E, subgroup 4	
Subgroup 5			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B	

TABLE IV. Groups B, C and E delta and electrical measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward - current transfer ratio	3076	I_C = 2.5 A dc, V_{CE} = 5 V dc, pulsed (see 4.5.1)	Δh_{FE2}	±20 percent change from initial reading		

1/ The delta measurements for table E-VIA (JANS) of MIL-PRF-19500 are as follows: Subgroups 4 and 5, see table IV herein, step 1.
2/ The delta measurements for 4.4.2.2 (JAN, JANTX and JANTXV) are as follows: All steps, see table IV herein, step 1.
3/ The delta measurements for table E-VII of MIL-PRF-19500 are as follows: Subgroup 6, see table IV herein, step 1.
4/ The delta measurements for 4.4.5 are as follows: Subgroups 1 and 2, see table IV herein, step 1.

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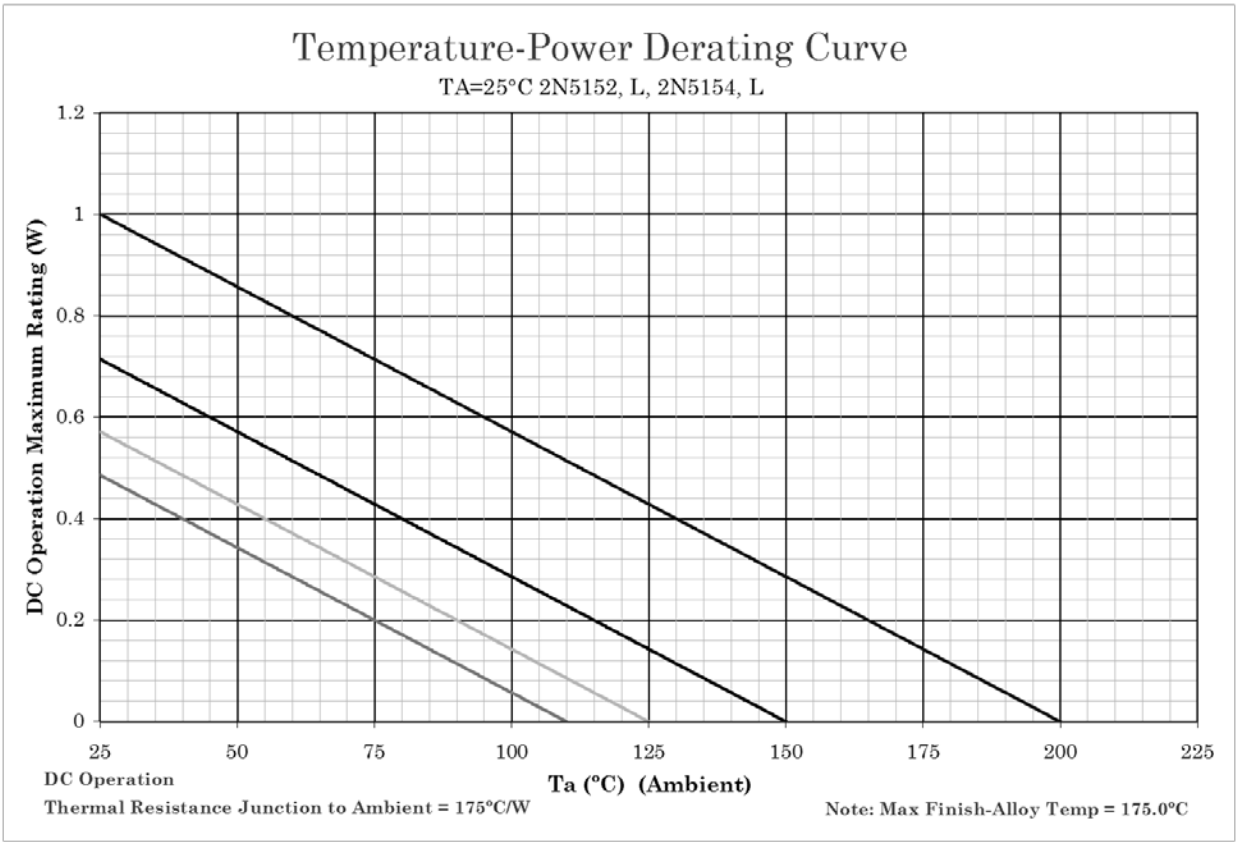


NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power/current for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 6. Temperature-power derating for 2N5152, 2N5152L, 2N5154, and 2N5154L, (TO-5 and TO-39).

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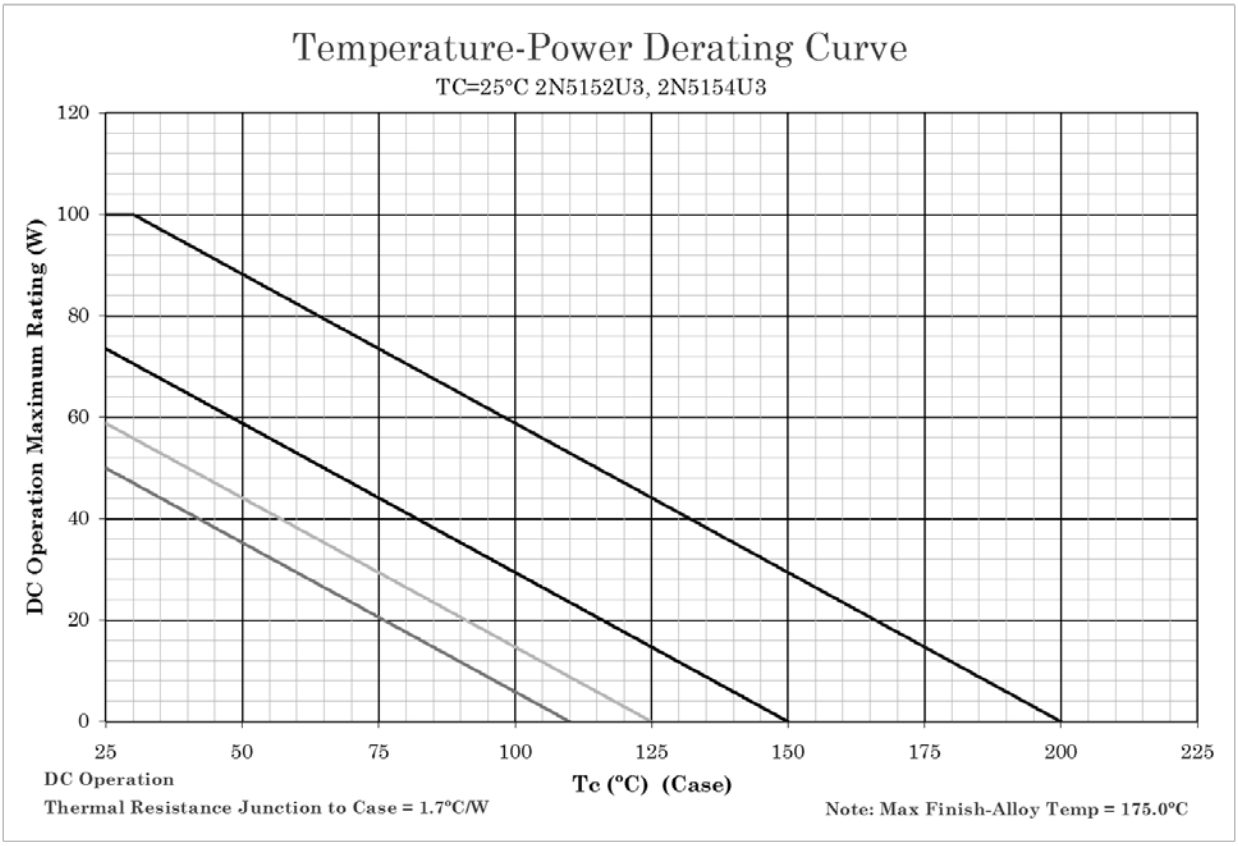


NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power/current for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 7. Temperature-power derating for 2N5152, 2N5152L, 2N5154, and 2N5154L, (TO-5 and TO-39).

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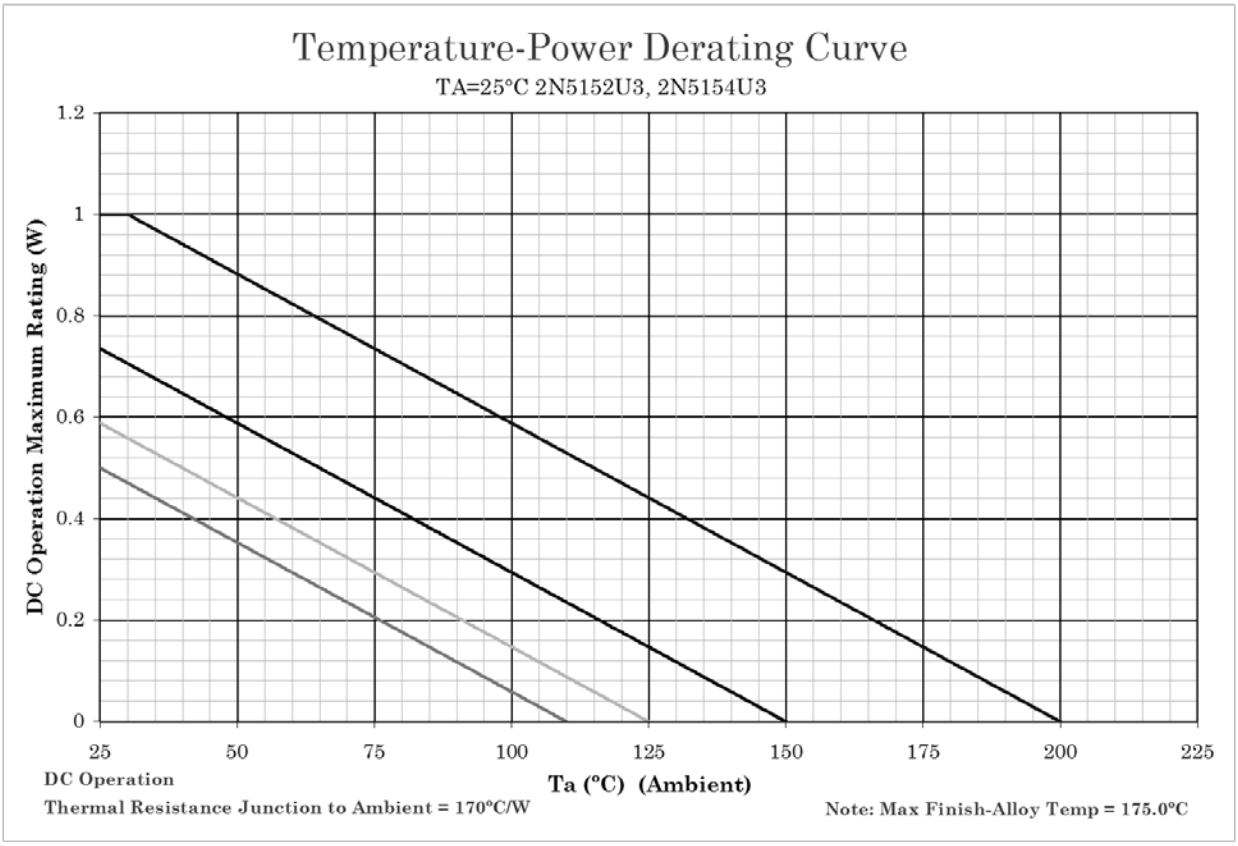


NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power/current for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 8. Temperature-power derating for 2N5152U3 and 2N5154U3.

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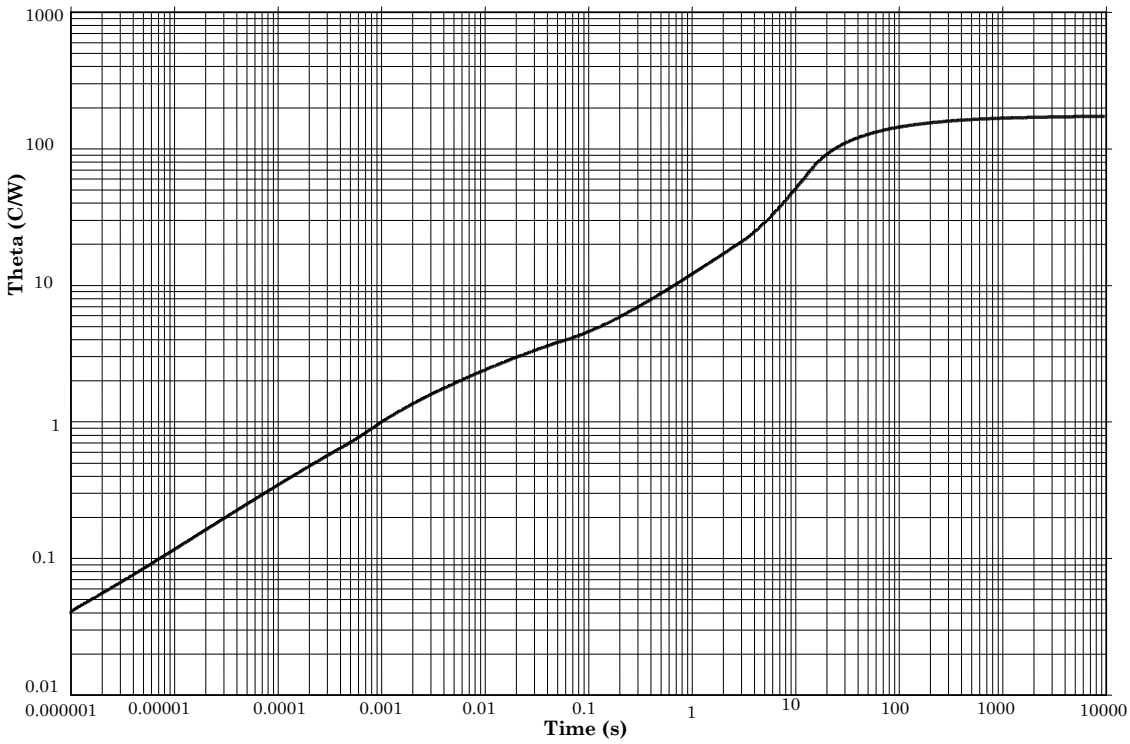
NOTES:

1. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power/current for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 9. Temperature-power derating for 2N5152U3 and 2N5154U3.

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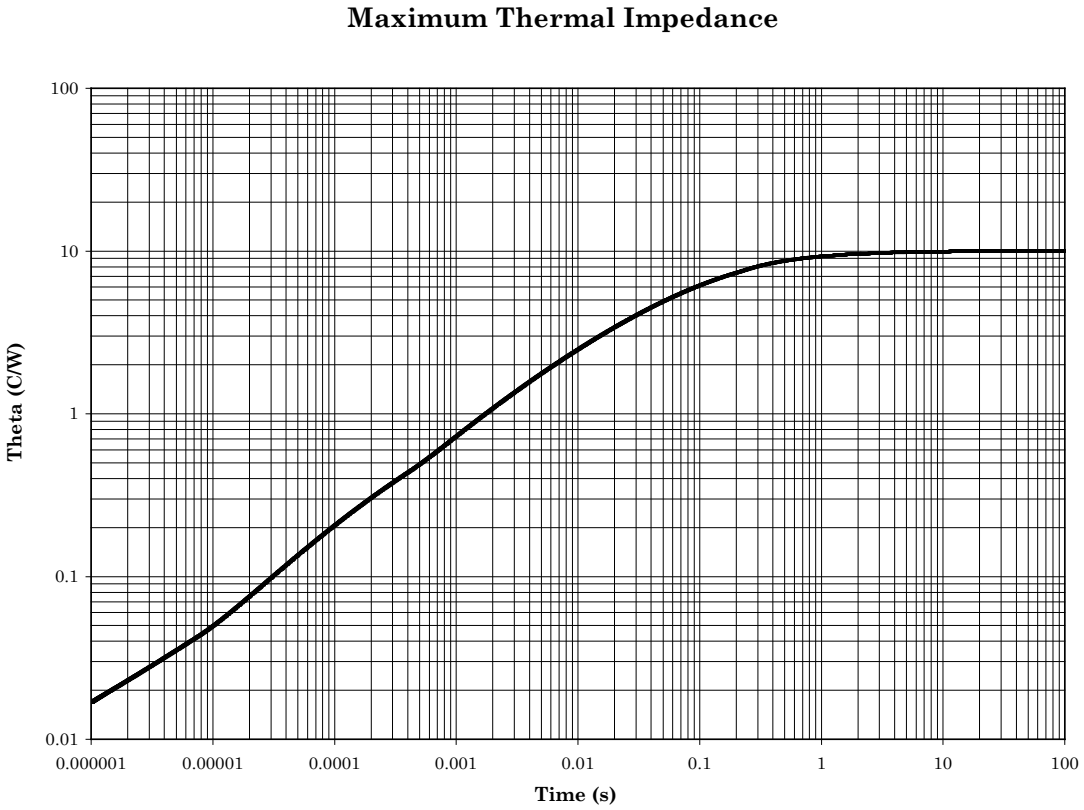
Maximum Thermal Impedance



T_A = +25°C, thermal resistance $R_{\theta JA}$ = 175°C/W.

FIGURE 10. Thermal impedance graph ($R_{\theta JA}$) for 2N5152, 2N5154, 2N5152L, and 2N5154L (TO-5 and TO-39).

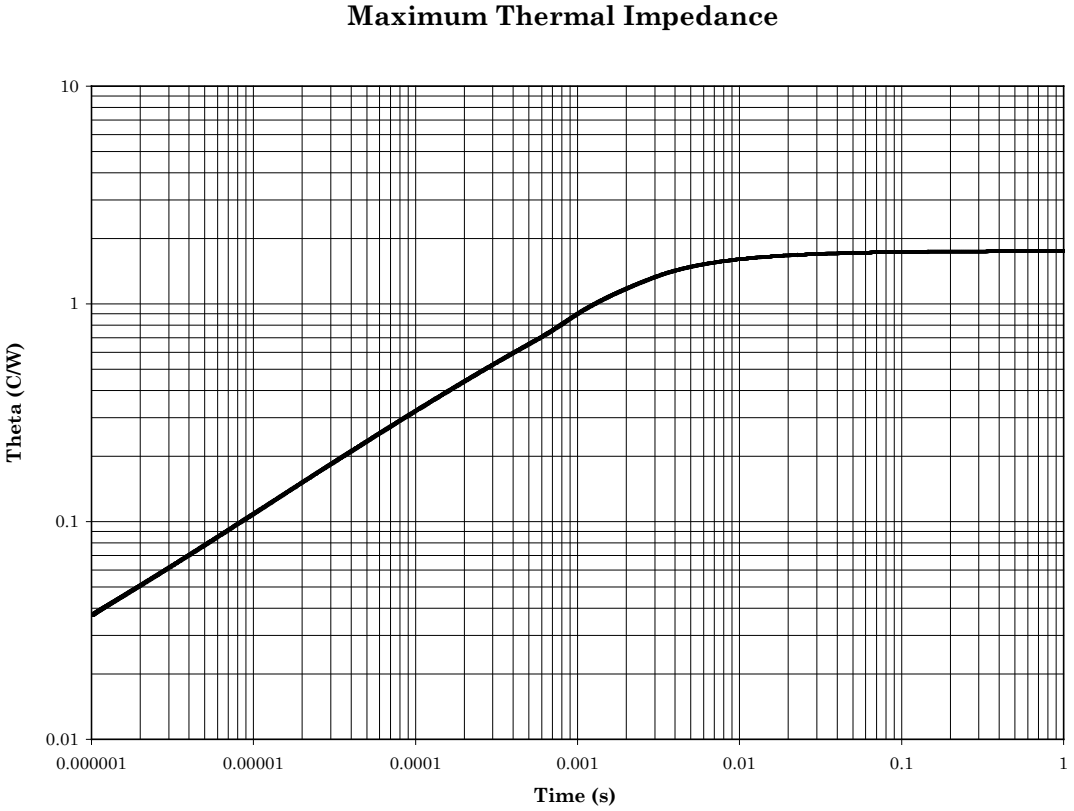
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Tc = +25°C, thermal resistance $R_{\theta JC}$ = 10°C/W.

FIGURE 11. Thermal impedance graph ($R_{\theta JC}$) for 2N5152, 2N5154, 2N5152L, and 2N5154L (TO-5 and TO-39).

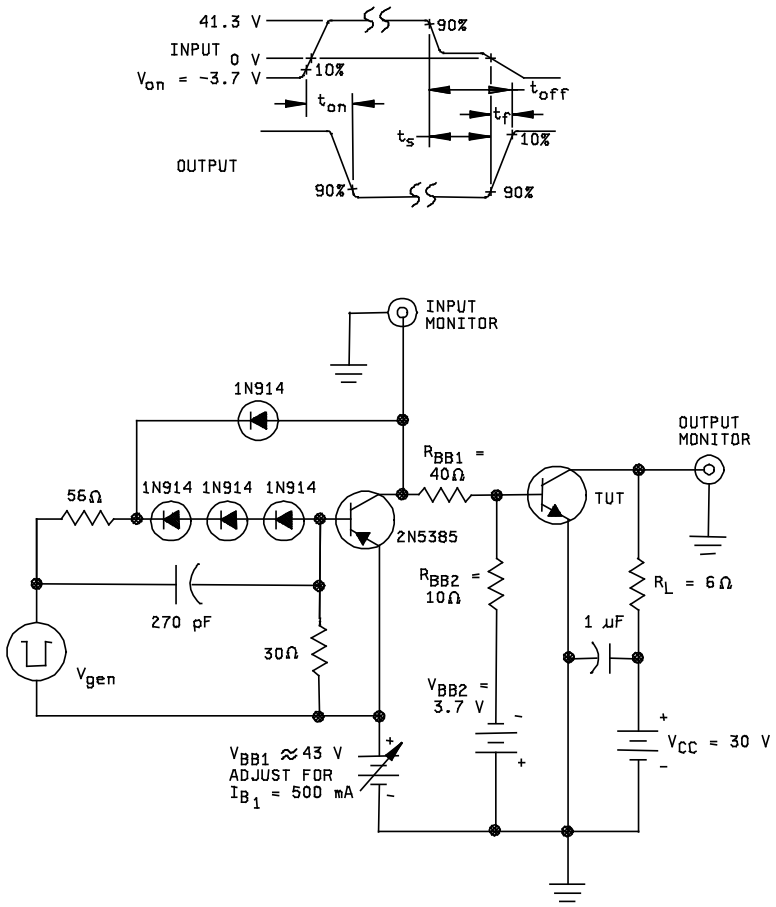
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T_C = +25°C. thermal resistance $R_{\theta JC}$ = 1.7°C/W

FIGURE 12. Thermal impedance graph ($R_{\theta JC}$) for 2N5152 and 2N5154 (U3).

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NOTES:

1. V_{gen} is a -30 pulse (from 0 V) into a 50 ohm termination.
2. The V_{gen} waveform is supplied by a generator with the following characteristics: $t_r \leq 15$ ns, $t_f \leq 15$ ns, Z_{out} = 50 ohm, duty cycle ≤ 2%, t_w = 20 µs.
3. Waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 1$ ns, $R_{in} \geq 10$ M ohm, $C_{in} \leq 11.5$ pF.
4. Resistors must be non-inductive types.
5. The dc power supplies may require additional bypassing in order to minimize ringing.
6. An equivalent drive circuit may be used.

FIGURE 13. Switching time test circuit.

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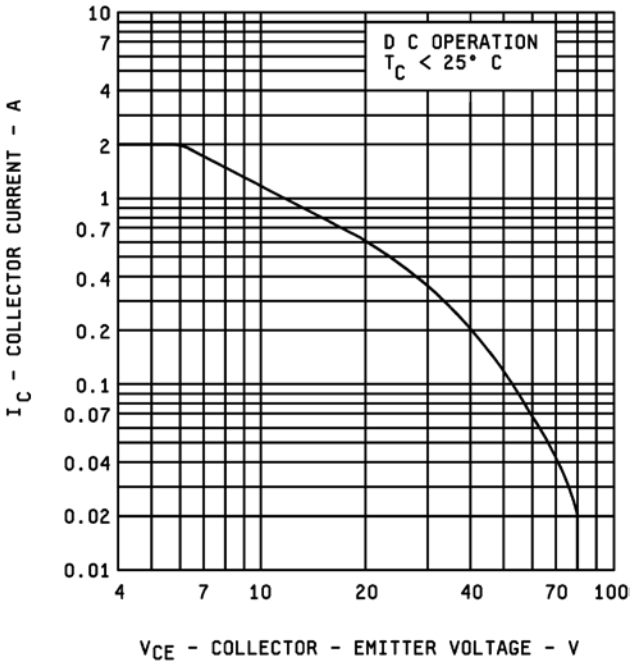


FIGURE 14. Maximum safe operating area.

R_{BB1} = 10Ω
R_{BB2} = 100Ω
L = 0.3 mH
R_L = 0.1Ω
V_{CC} = 10 V dc
I_{CM} = 10 A
V_{BB1} = 10 V dc
V_{BB2} = 4 V dc

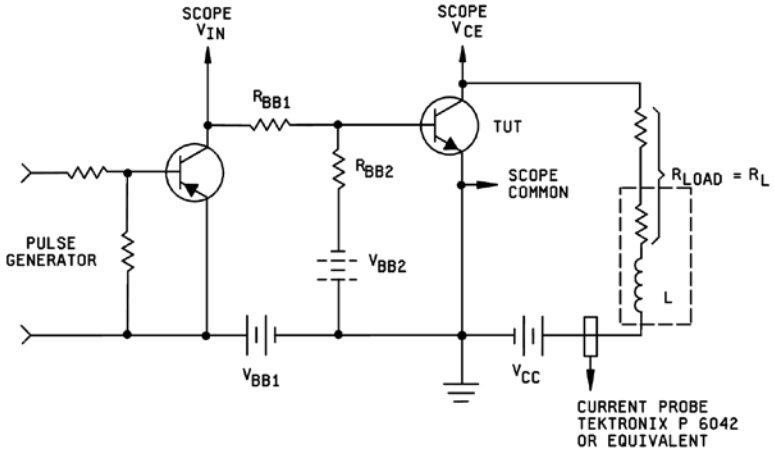


FIGURE 15. Unclamped inductive load energy test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.2).

* d. The complete Part or Identifying Number (PIN), see title and section 1.

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it must be specified in the contract.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

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* 6.4. Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCB2N5152) will be identified on the QML.

JANHC and JANKC ordering information			
PIN	Manufacturer		
	34156	43611	52GC4
2N5152 2N5154	JANHCB2N5152 JANHCB2N5154	JANHCD2N5152 JANHCD2N5154	JANHCE2N5152 JANHCE2N5154
2N5152	JANKCB2N5152 JANKCBF2N5152	JANKCD2N5152 JANKCDF2N5152	JANKCE2N5152
2N5154	JANKCB2N5154 JANKCBF2N5154	JANKCD2N5154 JANKCDF2N5154	JANKCE2N5154

* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activity:
Navy - MC
Air Force - 19, 71, 99

Preparing activity:
DLA - CC

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* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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